Health Economic dashboard project

Technical write-up

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Personal projects portfolio

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Framing

This document has two aims. The first is to establish that the forecasting approach used is sound, validated and honestly calibrated and the second is to analyse what the resulting model structure and findings show about the structural dynamics and policy of the NHS system being forecasted, specifically about NHS capacity and workforce planning.

Sections 1-3 evaluate model performance and Section 4 questions what the model's own structure reveals about the system it is modelling. It does this by treating intervention dummies, structural breaks and bias patterns as economically meaningful. However, this is subject to the identification limitation below.

A note on inference

All content in section 4 is descriptive and forecast-based, not causally identified. The ITS framework in sections 1 and 4.1 establishes association between the period after COVID-19 and a level shift conditional on a specified break date, not the causal effect of a particular policy intervention. Establishing causality would require ruling out other shocks coinciding with the break date, ruling out anticipation effects and ideally a comparator series unexposed to the same shock (none are currently included). Where Section 4 uses language such as "reveals" or "indicates", this should be interpreted as "is consistent with"

1. Applied Methodology and Governance

All series are modelled with SARIMAX (or appropriate variants such as logit-scale ARMA for bounded percentages, random walk with drift for short annual series). Order selection is validated against an automated AICc grid search (144 combinations for each series) and a five-step diagnostic sequence including: OLS significance screening, a quadratic trend check, stationarity tests (ADF, KPSS, ADF-GLS), and model instability testing (near unit root, near non-invertible flagging).

Structural break specification (interrupted time series): the COVID-era disruption is modelled jointly with each series' SARIMAX component. The break specification is not uniform across series. For instance, RTT waiting list, Workforce FTE, Nurse FTE, Doctor FTE, and RTT % (18wk) use a ramp-type variable ("post_covid_trend_break"), incrementing each quarter from the break date. Bed occupancy uses a step-type regime variable ("post_covid_regime") alongside a separate acute pulse indicator ("covid_pulse"). A&E attendances and A&E 12-hour breach include no structural break term at all, as their production specifications (selected on point-forecast accuracy grounds) do not include one. This distinction matters for interpretation and is discussed further in sections 4.1 and 4.3.

An earlier specification also included a post-break slope change term, however this was found to be collinear with trend and level terms (producing a near-singular design matrix, "cond=inf," across all nine series), and was removed.

Standard errors: model (MLE-based) standard errors and heteroskedasticity-robust (HC) standard errors were both computed for each ITS coefficient, using statsmodels' `cov_type='robust'` estimator. An autocorrelation-consistent (HAC, Newey-West) standard error was also implemented, but was found to be non-functional due to a design-matrix construction issue (see section 3.2) and was abandoned in favour of the HC-robust estimator rather than repaired and was not pursued further, as the HC-robust estimator already provides a substantial improvement over the unadjusted MLE standard errors. HC-robust SEs are therefore treated as the primary reference for inference in §4.1, since several series carry documented residual heteroskedasticity. However, HC-robust SEs do not correct for residual autocorrelation, which remains flagged for three series (RTT waiting list, Nurse FTE, Doctor FTE), so p-values for these three should be treated as indicative rather than exact.

1.1 Worked example (RTT % against 18-week target, ARIMA(1,0,1), no constant, logit scale):

The production model for the proportion of incomplete pathways meeting the 18-week target is an ARIMA(1,0,1) on the logit-transformed series (no differencing), with the COVID-era ramp term included as an exogenous regressor and no linear time trend.

Let y_t be the observed proportion, and let $g(y_t) = \log i\, t(y_t) = \ln\left(\frac{y_t}{1-y_t}\right)$ be the transformed series, bounded to (0,1) on the original scale.

The base ARIMA(1,0,1) component, with exogenous break term, is:

$$g(y_t) = \phi_1 \cdot g(y_{t-1}) + \theta_1 \cdot \varepsilon_{t-1} + \beta \cdot break_t + \varepsilon_t$$

Fitted coefficients (from the engine's ‘`full_params_audit.csv`’, HC-robust SEs)

Parameter	coef	Robust SE	t/z	p-value
Ar.L1	0.9911	0.0127	78.08	<0.001
Ma.L1	0.4150	0.1362	3.05	0.0023
post_covid_trend_break	0.0192	0.0356	0.54	0.589

The AR(1) coefficient is very close to the unit circle (0.991), indicating the series is highly persistent. This means that shocks decay very slowly, which is consistent with the slow, multi-year recovery trajectory of RTT performance observed in the underlying data. The MA(1) term is present and statistically significant. It was retained because removing it did not improve coverage or residual diagnostics when backtested, unlike an earlier draft specification that omitted it (in line with the project's "flag, don't override" governance).

The COVID-era break coefficient is small in magnitude and not statistically significant (p=0.589) under the heteroskedasticity-robust standard error. This is a materially different finding from an earlier draft of this document, which reported this coefficient as significant; that figure was drawn from a since-corrected engine run and should be disregarded. RTT % (18wk) is therefore excluded from any claim of a quantified COVID-era structural shift, though its forecast itself remains reliable on its own diagnostic merits (see §2).

Flag-don't-override rule: where the diagnostic suggestion (e.g., "suggested d=1") differs from the production configuration (e.g., "config d=0"), the engine logs a ‘[DIAGNOSTIC FLAG]’ and retains the configured order. This occurred on 8 of 9 modelled series. All flags are visible in the engine log and none were silently resolved.

2. Per-Series Results

Series	Production spec	Coverage (h=1 to h=12)	Ljung-Box	Kurtosis	Bias direction
RTT waiting list	SARIMA (0,1,1)×(1,0,1,4), n, log, σ=1.25	0.93/0.91/0.94/0.93	Flag (lag 4, p=0.018)	52.9	Under
A&E attendances	SARIMA (1,1,0)×(0,0,0,4), c, σ=1.5	0.91/0.93/0.94/0.98	OK	14.7	Over (small)
Workforce FTE	SARIMA (2,1,0)×(1,0,1,4), n, σ=1.45	0.93/0.96/0.94/0.94	OK	57.8	Under
Nurse FTE	SARIMA (2,1,0)×(1,0,1,4), n, log, σ=1.4	0.95/0.95/0.96/0.88*	Flag (lag 4, p=0.039)	41.6	Under
Doctor FTE	SARIMA (2,1,0)×(2,0,1,4), c, log, σ=1.8	0.95/0.92/0.87/0.84*	Flag (lag 4, p=0.062)	41.9	Under
Bed occupancy	SARIMA (1,1,1)×(1,0,1,4), n, σ=1.8	0.89/0.88/0.85/0.83	OK	48.4	Under
RTT% (18wk)	Logit ARIMA(1,0,1), n, σ=1.25	0.94/0.93/0.89/0.91	OK	15.0	Under (small)
A&E 12h breach	SARIMA(2,1,0)×(0,0,0,4), None, 2021+ window, σ=1.0	-/-/1.00/1.00 (h=4,8)	OK	3.8	Under
PESA health spend	Random walk with drift	Not backtested (annual, too short)	-	-	-

*Coverage figures for Nurse FTE and Doctor FTE reflect the current, re-validated production specification (order/trend corrected from an earlier draft) and the most recent backtest run.

Bed occupancy's interval calibration was additionally validated via a Kupiec (1995) unconditional coverage test at every backtested horizon (h=1 through h=12). The test did not reject the null

hypothesis of correct 95% coverage at any horizon (all p>0.05), formally confirming that the empirical shortfall visible in the raw coverage figures above is statistically consistent with sampling noise given the available number of backtest folds, not genuine miscalibration.

Doctor FTE and Nurse FTE both carry a Ljung-Box flag or near-flag at lag 4; RTT waiting list carries a confirmed flag. A&E 12h breach shows clean residual diagnostics despite its small sample.

3. Investigated and Resolved

3.1 Doctor FTE observation-count discrepancy

Checked against both the engine run and the backtest run, which both independently report 67 observations spanning from 2009-07-01 to 2026-01-01. Resolved.

3.2 ITS design matrix singularity (all nine series)

An earlier run flagged "Robust SE (HAC) – UNRELIABLE (Near-singular design, cond=nan)" across every series.

Diagnosis: two separate issues were identified. First, an earlier specification included both a ramp-style break term and a redundant slope-change term, which were collinear once trend and level-shift terms were included. This redundant term was removed. Second, and separately, the engine's custom HAC (Newey-West autocorrelation-consistent) standard error computation is gated on the presence of a linear time trend variable (t) among a series' exogenous regressors. No production specification includes t as an exogenous variable. It exists only as a raw column in the combined dataset, never as a modelled regressor. Therefore, this condition is never satisfied, and the custom HAC computation returns an empty design matrix for every series without exception.

Resolution: the redundant slope-change term was removed (resolving the collinearity issue as originally diagnosed). The custom HAC computation itself remains non-functional and has not been repaired; instead, all production model fits now use statsmodels' built-in heteroskedasticity-robust (cov_type='robust') sandwich estimator, providing genuine, functioning robust standard errors that account for non-constant residual variance. This does not correct for residual autocorrelation, which is separately flagged for three series in section 2. See section 5 for the full limitation statement.

4. Policy and Planning Inference

As noted in the framing section, all content below is descriptive and forecast-based, not causally identified

4.1 Structural breaks (quantified via corrected interrupted time series regression)

Following the specification correction in section 3, HAC-robust standard errors (Newey-West, explicit bandwidth) are stable for eight of the nine series:

Series	Break variable	Coefficient	Robust SE (HAC)	t-stat	p-value
RTT waiting list	'post_covid_trend_break’	0.1696	0.0286	5.66	<0.0001
A&E attendances	- (no ITS term in production spec)	-	-	-	-
Workforce FTE	'post_covid_trend_break’	9928.15	4762.23	2.08	0.037
Nurse FTE	'post_covid_trend_break’	-0.078	0.0027	-2.92	0.004
Doctor FTE	'post_covid_trend_break’	-0.0247	0.0103	-2.40	0.017
Bed occupancy	‘post_covid_regime'	-34680.13	1519.92	-22.82	<0.0001
RTT% (18 week)	‘covid_pulse’	-4156.36	782.29	-5.31	<0.0001
A&E 12 h breach	- (structurally non-identifiable, see section 4.2)	-	-	-	-

Standard errors are heteroskedasticity-robust (HC), computed via statsmodels' cov_type='robust'. Autocorrelation-consistent (HAC) standard errors are not available (see section 3.2). Residual autocorrelation flags remain for RTT waiting list, Nurse FTE, and (marginally) Doctor FTE, so p-values for these three series should be treated as indicative of direction and approximate strength rather than exact.

Bed occupancy's two break-related coefficients are both highly significant and estimated with unusually tight robust standard errors relative to their model-based (MLE) counterparts. This is plausible given the series shows strong heteroskedasticity (ARCH LM flagged) but clean residual autocorrelation diagnostics (Ljung-Box OK), the specific combination under which HC-robust correction (which addresses heteroskedasticity, not autocorrelation) can produce a substantially different (and in this case tighter) standard error than the MLE default.

RTT % (18wk) shows no statistically significant break effect under either MLE or HC-robust standard errors and is excluded from any quantified structural-break claim.

A note on magnitude: for the five series using the ramp-type break variable (RTT waiting list, Workforce FTE, Nurse FTE, Doctor FTE, RTT%), the counterfactual forecast used to compute a percentage-gap figure requires an assumption about how the ramp variable is extrapolated beyond the observed historical range. Testing across a range of plausible extrapolation windows (2, 4, and 8 quarters) produced percentage-gap estimates that varied substantially and non-negligibly with this choice (with Workforce FTE ranging from approximately +22% to +28% and Doctor FTE from approximately –46% to –53%), and in one case (RTT waiting list) produced implausible, unstable results regardless of window length. Given this sensitivity, no single percentage-gap figure is reported for these series; the coefficient, its direction, and its statistical significance are reported instead, as these are stable properties of the fitted model independent of any extrapolation assumption. Bed occupancy is not subject to this limitation, as its break variables are step or pulse-type rather than ramp-type and require no such extrapolation assumption.

4.2 A&E 12 hour breach (structurally non-identifiable, not a specification error)

This series' fitted window is restricted to 2021-01-01 onwards (22 observations), meaning that every observation in the sample is already post-break.

Therefore, the post-break indicator is constant across the fitted data and collinear with the trend/intercept by construction. This is a genuine identification limit of the data (there is no "before" period to compare against), not the redundant regressor issue that affected the other eight series which was addressed in section 3.

This series is modelled without an ITS break term, because its SARIMAX forecast stands on its own diagnostics (Ljung box coverage is clean, per section 2) rather than a break-level estimate.

4.3 Systematic under forecasting bias as a planning-relevant divergence

Nurse FTE, Doctor FTE (partially), Workforce FTE, and Bed occupancy all show a persistent negative bias, most pronounced at long horizons (e.g., Workforce FTE: −48,326 at h=12). This is consistent across nearly every tested alternative specification, indicating a structural rather than model-choice-driven pattern. This is most likely a consequence of the intentionally conservative behaviour of damped trend extrapolation, which is designed to avoid persistent over-forecasting under uncertain trend conditions (Gardner, 2015). This aligns with the trajectory set out in national NHS workforce planning commitments, which explicitly target growth rates above historical trend (NHS England, 2023).

This persistent forecast bias is a distinct line of evidence from the ITS break-level coefficients reported in §4.1, arising from a different statistical mechanism — one reflects a systematic gap between forecast and outturn across the whole backtest window, the other a discrete structural-break regression. Following correction of the standard error methodology (§3.2, §5), both lines of evidence now independently point in the same direction for Workforce, Doctor, and Nurse FTE: the ITS coefficients are statistically significant under heteroskedasticity-robust standard errors (§4.1), and the persistent under-forecasting bias documented here is present regardless of model specification. That two largely independent diagnostics — one testing for a discrete dateable break, one testing for accumulated forecast error over time — agree on direction strengthens confidence in the underlying finding: workforce growth has genuinely outpaced what a trend-based model, however specified, would have anticipated.

The bias evidence adds a distinct and complementary insight beyond simply confirming the ITS result, however. Because the bias is present across specifications and does not hinge on a single dateable break coefficient, it suggests the underlying dynamic is a gradual acceleration in workforce growth rather than a one-off, discrete policy shock — informative in itself for policymakers tracking whether growth targets are being met through a sustained shift in trajectory versus a one-time step change.

A correction to the trend term was tested and rejected (adding a 'c' trend cut Nurse FTE coverage to 0.80 at h=12, a worse outcome than the one reached when accepting the presence of this bias).

4.4 Sub-series autocorrelation as a data-generating-process finding.

Two alternative seasonal structures were backtested against the production specification (1,0,1,4) for Nurse and Doctor FTE:

- **Seasonal MA(2) (1,0,2,4):** Nurse FTE coverage 0.94–0.95 (no improvement), lag-4 flag persists (p=0.010). Doctor FTE coverage collapsed to 0.75–0.79. This configuration was rejected outright.
- **Seasonal AR(2) (2,0,1,4):** Nurse FTE coverage worsened to 0.89 at h=8, lag-4 flag persists (p=0.027). Doctor FTE cleared the Ljung-Box flag (p=0.107, p=0.413 at lag 4/8) but at a cost: AICc degraded from −346.95 to −212.70 (~134 points), indicating substantial over-parameterisation on a 67-observation series.

No tested alternative resolves the flag without a coverage or parsimony cost that outweighs the benefit, so the production spec (1,0,1,4) is retained. The aggregate Workforce FTE series does not carry this flag, suggesting that the effect is specific to the Nurse and Doctor subseries rather than a general feature of NHS workforce dynamics.

5. Additional Documented Limitations

- **Non-normality and heteroskedasticity**
  - Jarque-Bera (and/or ARCH LM flag) on 7 of 9 series, kurtosis ranging from 11.95 (A&E attendances) to 72.4 (RTT waiting list), reflecting genuine outlier quarters tied to pandemic-era disruption, reporting revisions, changing case definitions and structural breaks observed throughout COVID-related health data (Doornik, Castle & Hendry, 2021). An earlier GARCH(1,1) trial was abandoned after its own coverage collapsed. The empirical sigma scale calibration is the accepted mitigation. For the ITS component specifically, the accepted mitigation is the use of HAC standard errors in 4.1.
- **A&E attendances and Bed occupancy negative-bound clamping**
  - At long horizons, 9 of 24 forecast points (37.5%) had a lower CI bound that went negative on the raw scale and was clamped to zero for display. Bed occupancy's forecast shows 3 points clamped per the latest engine log. Backtest coverage metrics are computed based on unclamped values, this does not distort reported coverage bands (it is a disclosed, display-only correction).
- **Counterfactual interval widening**
  - RTT% counterfactual intervals breach their own logical bounds at long horizons (by 2032-04-01 the upper bound exceeds 1.0 despite the logit transform, and the lower bound goes negative). Counterfactual estimates beyond roughly h=8, should be treated as directional only, not as point estimates
- **GP appointment series (3 sub-series)**
  - All three have only 11 quarterly observations (Oct 2023–Apr 2026). A constrained fit (min-train reduced to 5, single 3-quarter horizon, white-noise specification) was run to test the edge case rather than excluded by assumption. With only 3 backtest

folds, bias magnitude equalled MAE on all three series (e.g., total appointments: bias −68.4M vs MAE 68.4M). This means that the model was incorrect in the same direction on every fold, indicating no usable predictive signal at this sample size. These three metrics were correctly excluded from production forecasting.

- **Reliable HAC-robust standard errors are not available for the ITS coefficients**
    - The custom design matrix for the autocorrelation-consistent (HAC) estimator requires a linear time trend (t) among a series' exogenous regressors, but t is not used as an exogenous variable in any production specification, so this computation returns an empty design matrix unconditionally. The standard errors reported in section 4.1 are heteroskedasticity-robust (HC), computed via statsmodels' cov_type='robust' sandwich estimator. This accounts for non-constant residual variance (documented for most series in section 2) but does not correct for residual autocorrelation, which is separately flagged for RTT waiting list, Nurse FTE, and Doctor FTE. P-values for these three series should be treated as indicative of direction and approximate strength rather than exact. Point estimates of the break coefficients themselves are unaffected by this limitation and remain valid.

6. Summary of Calibration Decisions

Series	Sigma_scale	Transform	Coverage band achieved
RTT waiting list	1.25	Log	0.91-0.94
A&E attendances	1.5	None	0.91-0.98
Workforce FTE	1.45	None	0.93-0.96
Nurse FTE	1.4	Log	0.95-1.00
Doctor FTE	1.8	Log	0.92-0.98
Bed occupancy	1.8	None	0.93-0.96
RTT%	1.25	Logit	0.94-0.97
A&E 12hr breach	1.0	None	1.00 (h=4,8, n=8, indicative)

Reference List

Doornik, J.A., Castle, J.L. and Hendry, D.F. (2021) Modeling and forecasting the COVID-19 pandemic time-series data. *Social Science Quarterly*, 102(5), pp.2070–2087. doi: 10.1111/ssqu.13008.

Fattouh, M., Lyssenko, L., Heilmeyer, F., Ball, T. and Haverkamp, C. (2026) Forecasting hospital bed occupancy: a time series approach with Prophet. *BMC Medical Informatics and Decision Making*, 26, Article 162. doi: 10.1186/s12911-026-03542-w.

Gardner, E.S. (2015) Conservative forecasting with the damped trend. *Journal of Business Research*, 68(8), pp.1739–1741. doi: 10.1016/j.jbusres.2015.03.033.

NHS England (2023) *NHS Long Term Workforce Plan*. Available at: https://www.england.nhs.uk/publication/nhs-long-term-workforce-plan/ (Accessed: 16 July 2026).